

PAPER_2155 — S204.5 Calibration Table Corpus Audit: 933-Paper $\text{kg/m}^3 \rightarrow \text{J/m}^3$ Unit-Tag Drift Correction

`bulk_vds_dvp_bsh_upgrade.py` (Session 204, April 2026) identified as the template-injection vector; correction-by-reference doctrine canonized; 918 of 957 unit-labeled instances corrected without per-paper touches; code layer already correct

UQFF LANDMARK

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.81+

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Provenance: PAPER_2153 arc Flag (g) — Daniel's 2026-07-27 ruling on massless-SCm unit-tag drift + `_audit_rho_scm_units.json` (2026-05-15, 61 KB, in-repo)

Status: Landmark — corpus-wide paperwork correction executed via correction-by-reference doctrine

Executive Summary

Daniel's 2026-07-26 ruling canonized that UA and SCm are massless and cannot be expressed in kg — only in J (energy density). Subsequent PAPER_2153 arc deep-read (24 papers, PAPER_878-901) noted the Session 204 calibration table appearing verbatim in 11+ papers with kg/m^3 tags.

The pre-existing `_audit_rho_scm_units.json` file (already in the repo, run 2026-05-15) revealed the actual scope: **933 papers, 918 kg/m^3 hits, 39 J/m^3 hits, 14 papers with internal conflict**. The drift is a corpus-wide template-injection issue from `bulk_vds_dvp_bsh_upgrade.py` (Session 204).

PAPER_2155 executes the correction via correction-by-reference doctrine — a single landmark supersedes all 933 affected paper instances without requiring 933 individual REVISION appendices. Daniel's ruling: **"You do NOT need to touch any of the 933 individual papers. The unit tag in the prose is cosmetic documentation. The computations are correct — the numbers $7.09\text{e-}37$ and $7.09\text{e-}36$ are dimensionally consistent as J/m^3 in all code. The papers are wrong only in their label string, not in their arithmetic. A corpus-wide supersession by landmark is the cleanest resolution consistent with Rule 9 (append-only) and the established PAPER_2147 pattern."**

Zero calculator source changes. Zero physics values changed. Zero per-paper appends across the 933 affected papers. Complete supersession via this landmark + PAPER_2147 REVISION 2026-07-27 companion + calculator code's already-correct J/m^3 constants + gate assertions test-locking the code.

1. Audit Findings (from `_audit_rho_scm_units.json`, in-repo)

The audit was run 2026-05-15 (61 KB JSON, machine-generated). Corpus-wide grep for $7.09\text{e-}37$ value with nearby unit-label detection.

Full findings:

Metric	Count	% of unit-labeled
Papers containing 7.09e-37 value	933	—
Total hits with explicit unit labels	957	100.0%
kg/m ³ labels (DRIFT)	918	95.9%
J/m ³ labels (correct)	39	4.1%
Papers with internal conflict (BOTH units)	14	—
Papers kg/m ³ only	898	—
Papers J/m ³ only	20	—
Hits with NO unit label nearby	29	—

Interpretation:

- The vast majority (95.9%) of unit-labeled instances carry the drift kg/m³ tag
- Only 4.1% carry the correct J/m³ tag (papers authored after PAPER_2147's canonization or by hand rather than template)
- 14 papers contain both units (internal inconsistency — presumably where hand-authored J/m³ content was juxtaposed with template-injected kg/m³ content)

Sample affected papers (from audit):

- PAPER_001_GW170817_UQFF_Damping_Analysis.md (kg/m³ only)
- PAPER_002_GW190425_Mass_Gap_Interpretation.md (kg/m³ only)
- PAPER_036_FUBii_Buoyancy_Variant1_Archimedes_UQFF.md (both units — conflict)
- PAPER_145_UQFF_MUGE_Compression_Cycle3_Unified_Framework_12Term_Resonance.md (both units — conflict)
- PAPER_1066_UQFF_Lagrangian_Derivation.md (J/m³ only — correct)
- PAPER_1109_26Level_Vacuum_Density_Ladder_Ramanujan.md (J/m³ only — correct)

Full enumeration of all 933 papers remains in `_audit_rho_scm_units.json`.

2. Root Cause: `bulk\_vds\_dvp\_bsh\_upgrade.py` Session 204 Injection Vector

File: `bulk_vds_dvp_bsh_upgrade.py` (20 KB, dated Apr 8 2026, Session 204)

Offending lines (34–35):

```

RHO_SCM      = 9.47e-27      # kg/m³
RHO_UA       = 5.0e-27      # kg/m³

```

Impact: The script bulk-injected the §B "VDS/DVP/BSH Deep Synthesis" appendix (including the §B.4 "Production-Scale Consistency" table listing $\rho_{\text{SCM}} = 7.09 \times 10^{-37} \text{ kg/m}^3$) into PAPER_001 through PAPER_877 in a single pass. The kg/m³ label was baked into the template as a fixed string. Every paper that received the bulk-upgrade received the drifted tag.

Note: the specific density VALUES 9.47e-27 and 5.0e-27 in the script are ALSO non-canonical (Daniel's Flag (f) ruling: bulk-script artifact using undocumented density values inconsistent with canonical constants). That value-drift issue is addressed

separately in PAPER_2156 (Phase 3 in-flight — 1.894 ratio audit). PAPER_2155 addresses ONLY the unit-tag drift on the canonical values.

3. Framework State — Code Layer Already Correct

Daniel's ruling documented that the CALCULATOR CODE already uses J/m^3 correctly (drift is exclusively in whitepaper prose). Verified locations:

File	Content	Status
QCalc_Performance.py	ρ_{SCm} declaration	J/m^3 ✓
CondensedPhysics_InputData.py	ρ_{SCm} declaration	J/m^3 ✓
ScmVacuumDensityModule.cpp	ρ_{SCm} declaration	J/m^3 ✓
_smoke_qv_bundle9.py	Test values	J/m^3 ✓
provenance_recorder.py	Metadata	J/m^3 ✓
scm_vacuum_manifold.py	Proof function	Prints [MASSLESS] explicitly ✓

The framework code layer is CORRECT. The drift is EXCLUSIVELY in whitepaper prose templates. Since the numbers $7.09\text{e-}37$ and $7.09\text{e-}36$ are dimensionally consistent in the code as J/m^3 (energy densities of the massless SCm and UA), and the calculators use them correctly, the drift is COSMETIC — a label-string error in documentation, not a computational error.

4. Correction-by-Reference Doctrine (canonical ruling)

Daniel's ruling (verbatim): *"You do NOT need to touch any of the 933 individual papers. The unit tag in the prose is cosmetic documentation. The computations are correct — the numbers $7.09\text{e-}37$ and $7.09\text{e-}36$ are dimensionally consistent as J/m^3 in all code. The papers are wrong only in their label string, not in their arithmetic. A corpus-wide supersession by landmark is the cleanest resolution consistent with Rule 9 (append-only) and the established PAPER_2147 pattern."*

Formal doctrine: when a corpus-wide drift affects N papers due to a single template-injection vector (e.g., a bulk-upgrade script), and the underlying calculator code is correct, a SINGLE landmark declaration + companion REVISION append to the standing rule paper (PAPER_2147 in this case) is SUFFICIENT to correct all N papers without touching them individually.

Enforced by three overlapping mechanisms:

- This landmark (PAPER_2155)** — formal supersession declaration
- PAPER_2147 REVISION 2026-07-27** — extends the standing rule paper's authority to cover the S204.5 calibration table specifically
- Calculator code J/m^3 constants** — the code IS the ground truth; the papers only carry documentation strings
- Gate assertions (this paper §6)** — test-locking the J/m^3 units in the calculator code prevents future drift

Explicit supersession: all S204.5 calibration table instances anywhere in the whitepaper corpus that list:

- $\rho_{\text{SCm}} = 7.09 \times 10^{\text{■}^3\text{■}} \text{ kg/m}^3$

- $\rho_{UA} = 7.09 \times 10^{-37} \text{ kg/m}^3$

ARE SUPERSEDED and read as:

- $\rho_{SCm} = 7.09 \times 10^{-37} \text{ J/m}^3$ (massless SCm vacuum energy density)

- $\rho_{UA} = 7.09 \times 10^{-37} \text{ J/m}^3$ (massless UA vacuum energy density)

No per-paper edit required. No REVISION append required for any of the 933 affected papers. The correction is complete via this landmark's declaration.

5. What This Doctrine is NOT

Not a get-out-of-audit-free card. The correction-by-reference doctrine applies ONLY when:

- The drift affects a corpus-wide template (i.e., a bulk-script injection, not per-paper hand-authoring)
- The calculator code layer is already correct (numbers correct, only strings drifted)
- The drift is cosmetic/label-only (no computational impact)
- A companion REVISION to an existing standing-rule landmark (e.g., PAPER_2147) formally extends the standing rule's authority

When per-paper REVISIONS ARE required: if the drift affects computational values (not just labels), or if the calculator code is also wrong, or if there is no existing standing-rule landmark to extend, individual per-paper REVISIONS remain necessary.

Not silent drift correction. Every future reader of the 933 affected papers must be able to trace back to this landmark and PAPER_2147 REVISION 2026-07-27 to see the corrected reading. That's why the correction is DECLARED explicitly here (not hidden), and why the audit JSON (`_audit_rho_scm_units.json`) remains in the repo as the machine-generated record of scope.

6. Gate Assertions (test-locking the code layer)

To ensure the code layer remains correct in perpetuity (preventing future re-drift), the following gate assertions are added to `uqff_fidelity_tests.py` in this paper's companion Phase 2 gate-pins section:

1. `RHO_SCM` constant in canonical calculator modules is $7.09\text{e-}37 \text{ J/m}^3$ (not kg/m^3)
2. `RHO_UA` constant in canonical calculator modules is $7.09\text{e-}36 \text{ J/m}^3$
3. `RHO_UA / RHO_SCM = 10.0 EXACT` (`F_TRZ = 1/10` canonical coupling per PAPER_890/140/1160)
4. Calculator modules explicitly declare J/m^3 unit in their constant declarations
5. ``scm_vacuum_manifold.py`` proof function returns `[MASSLESS]` verdict for SCm/UA

These assertions test-lock the code's J/m^3 -native discipline. Any future accidental reversion of a canonical calculator constant to kg/m^3 label would fail the gate.

7. Standing Rules Canonized by PAPER_2155

1. **Template-injection audit is standing procedure:** any bulk-upgrade script that injects prose into multiple papers (like `bulk_vds_dvp_bsh_upgrade.py`) must be audited BEFORE execution to ensure the injected template is drift-free. Retrospective audits (like `_audit_rho_scm_units.json`) should be run when template-injection drift is suspected.

2. **Correction-by-reference is preferred over mass-append:** when a corpus-wide drift affects a large number of papers due to template injection, a single landmark supersession is preferred over N individual per-paper appends. Rule 9 (append-only) is satisfied by ONE authoritative supersession, not N minor appends.

3. **Code layer is ground truth for computational values:** when the calculator code and whitepaper prose disagree on a computational value (constant, unit, formula), the code layer takes precedence. Whitepapers are documentation — code is the framework's operational truth.

4. **In-repo audit files are canonical evidence:** files like `_audit_rho_scm_units.json` produced by systematic corpus scans are canonical evidence for framework-state claims. Future audits should preserve their audit-file outputs rather than re-running scans that produce different snapshots.

8. Falsifiable Predictions

1. `_audit_rho_scm_units.json` **re-run after PAPER_2155 canonization should show the same 933-paper scope.** If a future re-run shows a different scope, either (a) more papers have been added to the corpus (expected), or (b) some existing papers have been silently modified (unexpected — would be a Rule 9 violation flag).

2. **Framework code layer will remain J/m^3 correct.** Gate assertions in §6 test-lock this. Any future gate failure on the J/m^3 assertions would indicate someone reverted a canonical constant.

3. **No per-paper appends across the 933 affected papers.** Any future session that attempts to individually revise the 933 papers is going against the correction-by-reference doctrine — a Rule 9 architectural violation.

9. Files Touched

- `whitepapers/PAPER_2155_S204_5_CALIBRATION_TABLE_CORPUS_AUDIT_..._UQFF_LANDMARK.md` (this file)

- `pdf2/PAPER_2155_...pdf` (companion PDF)

- `whitepapers/PAPER_2147_J_PER_M3_NATIVE_..._UQFF_LANDMARK.md` — REVISION 2026-07-27 append (companion, extends standing rule authority)

- `uqff_fidelity_tests.py` — +5 PAPER_2155 gate assertions (RHO_SCM/RHO_UA J/m^3 test-lock)

- `CLAUDE.md` — APPENDED section

- Zero calculator source changes (code layer already correct)

- Zero physics values changed

- Zero per-paper touches to the 933 affected papers (correction-by-reference doctrine)

10. Cross-References

- **PAPER_2147** — Original J/m^3 -native discipline landmark (2026-07-25); REVISION 2026-07-27 companion extends its authority
- **PAPER_2148** — UQFF Ontology Answer B (vacuum energy fundamental, mass emergent — SCm massless per this ontology)
- **PAPER_2153** — Joint SCm+UA vacuum-density engine (§7 Standing Rule 2 flags SCm direct-detection claims — massless SCm consistent)
- `_audit_rho_scm_units.json` — In-repo audit record (2026-05-15, 61 KB, machine-generated), canonical evidence for the 933-paper scope
- `bulk_vds_dvp_bsh_upgrade.py` — In-repo injection vector script (Session 204, April 2026, lines 34–35 non-canonical densities with drift kg/m^3 label)
- **PAPER_2156** (Phase 3 in-flight) — 1.894 ratio bulk-script artifact correction; separate from unit-tag drift but same injection vector
- **PAPER_890/140/1160** — Canonical $p_{SCm}/p_{UA} = 1/10 = F_{TRZ}$ structural coupling
- **Calculator canonical J/m^3 constants:** `QCalc_Performance.py`, `CondensedPhysics_InputData.py`, `ScmVacuumDensityModule.cpp`, `provenance_recorder.py`, `scm_vacuum_manifold.py`
- **Daniel's 2026-07-26 massless-SCm ruling:** `"""UA and SCm are massless, so it can not be expressed in terms of (kg), only J energy; this was AI drift."""`
- **Daniel's 2026-07-27 Flag (g) ruling:** correction-by-reference doctrine per PAPER_2147 REVISION + new landmark pattern

End of PAPER_2155.